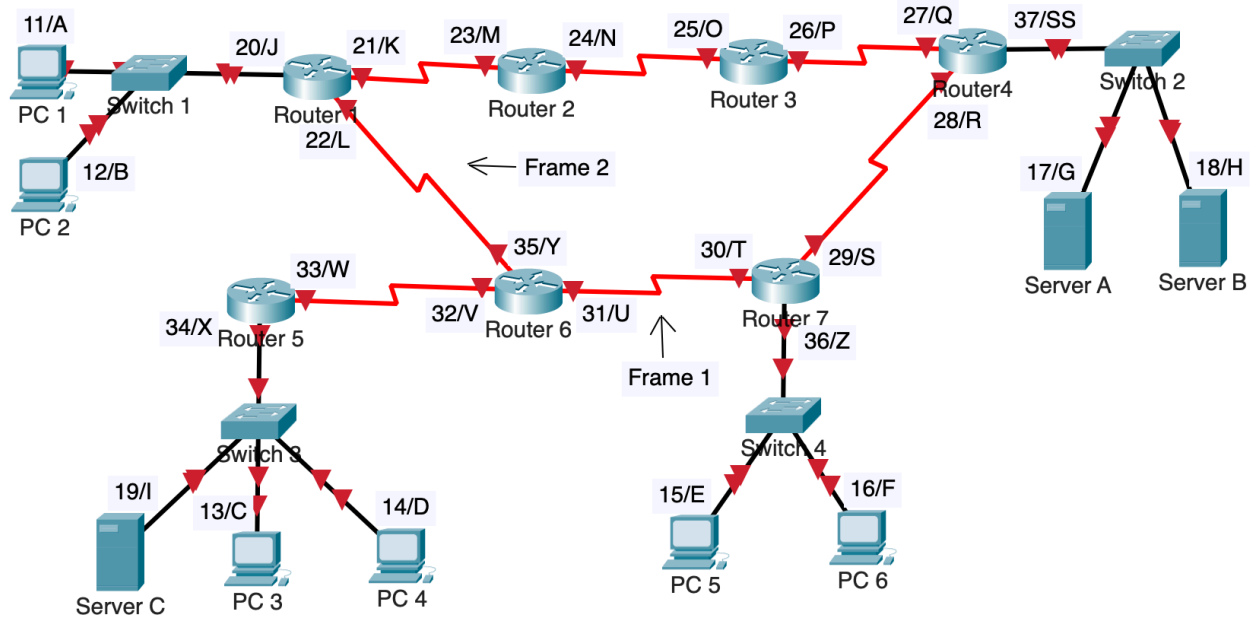


Name:		ID:		Section:	
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1. For the following network configuration, assume the physical addresses to be the alphabets and logical addresses to be the numbers.



- PC4 is downloading **Frame 1** from File Server A which is using port 20.
- Frame 2** is coming from PC2 and it is intended for Email Server C which is using port 25.

Complete the following frame headers using appropriate source and destination addresses. [5]

Frame1	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port
	U	T	14	17	50001	20
Frame2	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port
	Y	L	19	12	25	50002

2. Write which layer in the **OSI model** is responsible for the following data communication tasks. [3]

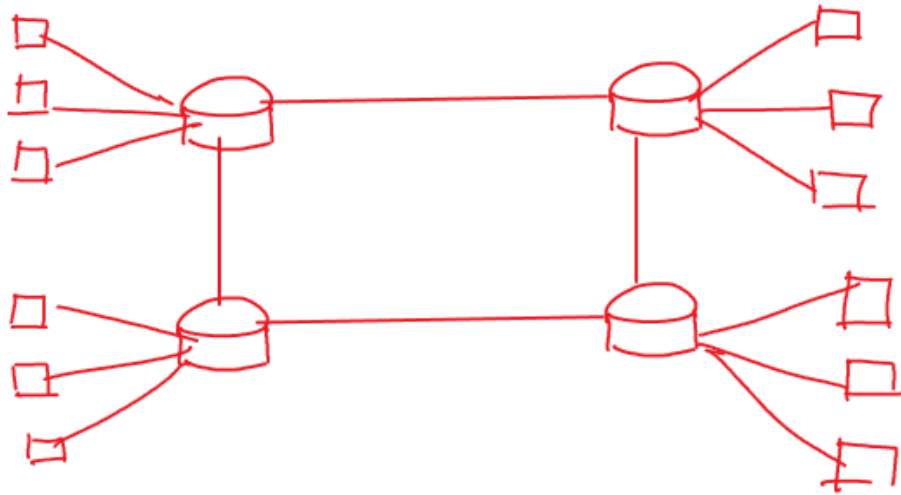
- Finding out the IP address of youtube using DNS service. **Application layer**
- Flow and error control in a link between two routers. **Data-link layer**
- Auto-authentication using saved credentials for youtube. **Session layer**

3. Explain **buffering** using a real life example. [2]

Solution: Slow or inconsistent arrival of data in recorded communication, such as youtube video or netflix movies.

4. Consider a hybrid wired network topology with **FOUR** routers, where each router is connected with its two neighbouring routers only. **THREE** end-devices are connected with each central router by using a point-to-point connection. Each wired link costs 10 taka.

- A. Draw the hybrid network structure. [2]
- B. Calculate the total cost for setting up the network configuration. [2]
- C. Identify a disadvantage in the topology used for the end devices. [1]

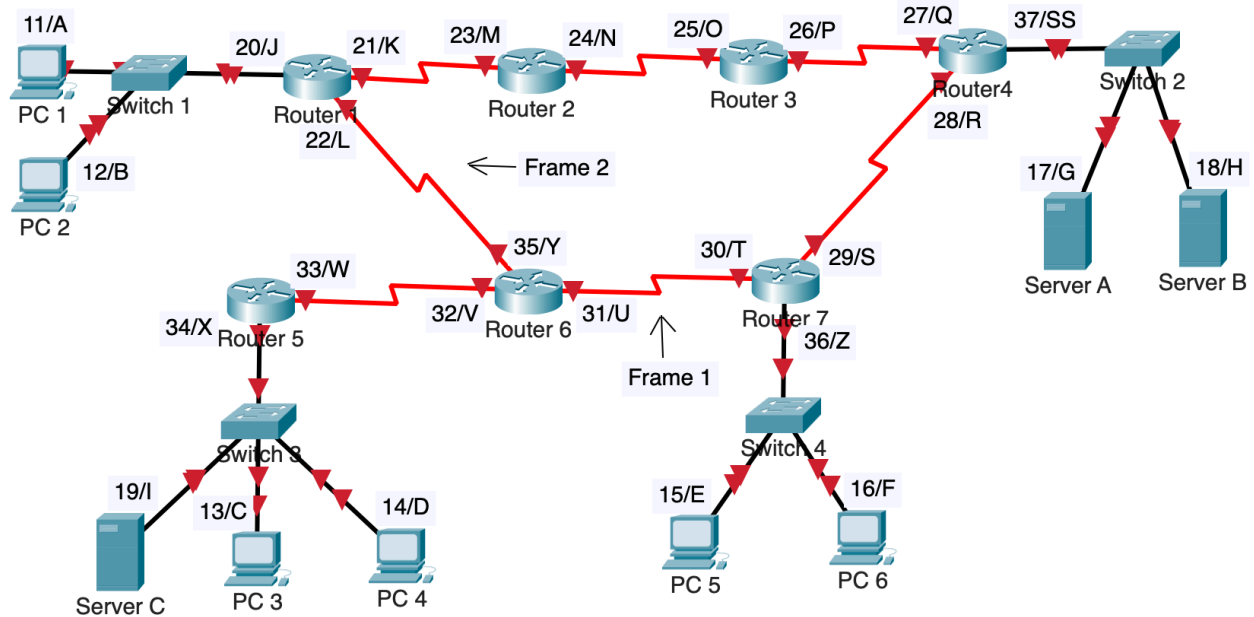


$$4 + 4 \times 3 = 16 \rightarrow 160 + k$$

Topology used for end devices – Star. Disadvantage: Single point of failure.

Name:		ID:		Section:	
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1. For the following network configuration, assume the physical addresses to be the alphabets and logical addresses to be the numbers.



- PC3 is downloading **Frame 1** from File Server B which is using port 25.
- Frame 2** is coming from PC1 and it is intended for Web Server C which is using port 80.

Complete the following frame headers using appropriate source and destination addresses. [5]

Frame1	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port
	U	T	13	18	50001	25
Frame2	D. Mac	S. Mac	D. IP	S. IP	D. Port	S. Port
	Y	L	19	11	80	50002

2. Write which layer in the **OSI model** is responsible for the following data communication tasks. [3]

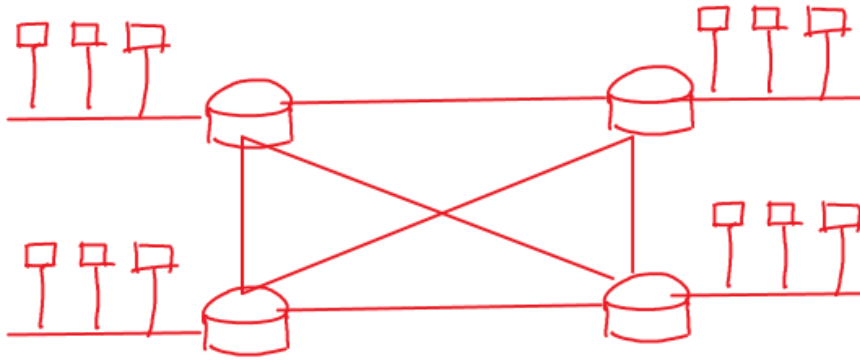
- Translating the images and files attached with an email. **Presentation layer**
- Writing the physical address of the next hop. **Data-link layer**
- Logging a user out of the email application after a certain time. **Session layer**

3. Explain **jittering** using a real life example. [2]

Solution: Slow or inconsistent arrival of data in live or real-time communication, such as live football match streaming or zoom calls.

4. Consider a hybrid wired network topology with **FOUR** routers, where all routers are connected with one another. **THREE** end-devices are connected with each router by using a multi-point connection. Each wired link costs 12 taka.

- A. Draw the hybrid network structure. [2]
- B. Calculate the total cost for setting up the network configuration. [2]
- C. Identify a disadvantage in the topology used for connecting the routers. [1]



$$\frac{4 \cdot 3}{2} + 4(3+1) = 22 \rightarrow 220 \text{ Tk}$$

Topology used for connection routers – MESH. Disadvantage: Costly, too many wires needed.